Advanced Machine learning Mastering Course

Introduced by

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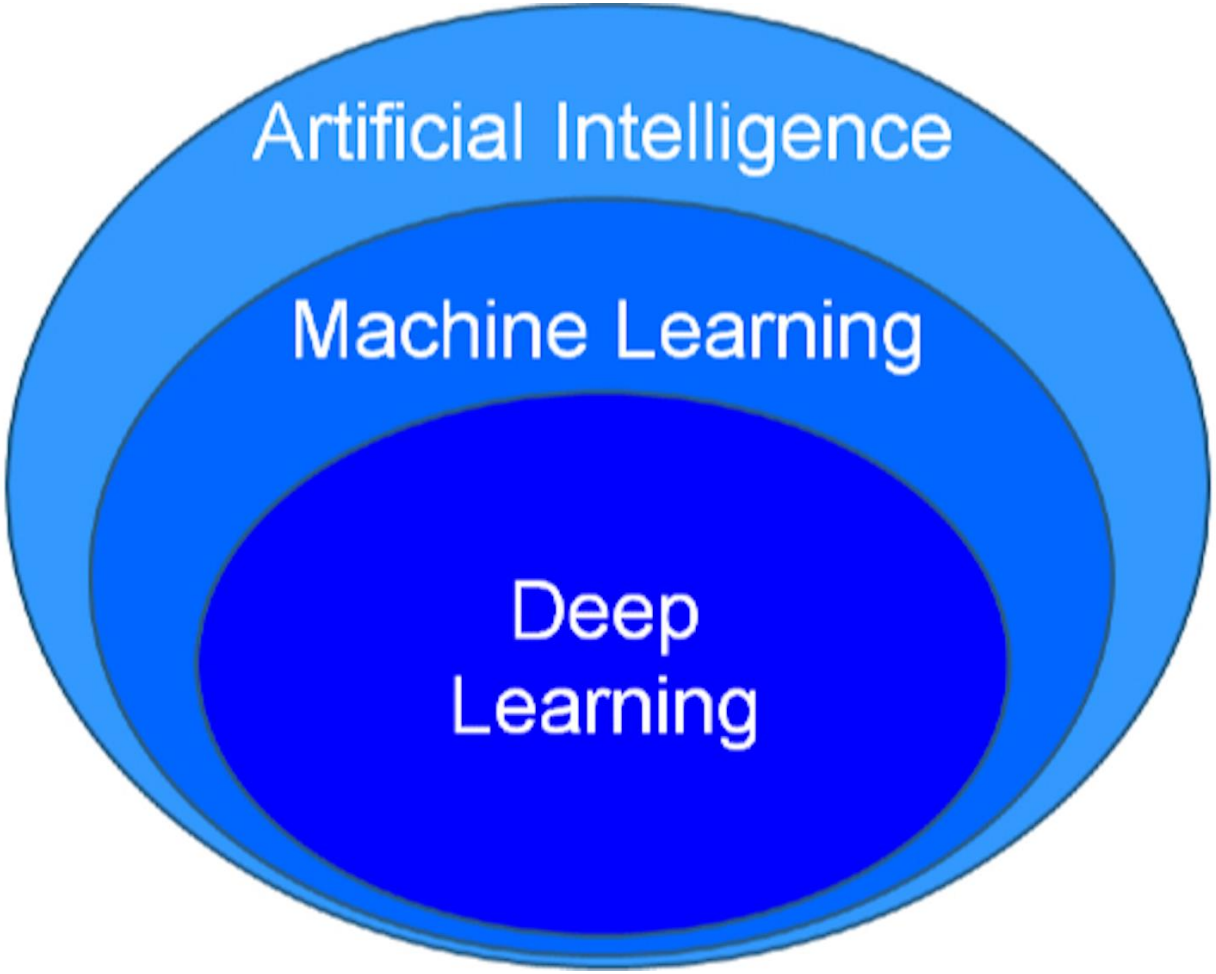
Agenda

1. Machine Learning Foundations
2. Training Models
3. Testing Models
4. Evaluation Metrics
5. Detecting Errors
6. Putting it all together
7. Bayesian Learning
8. Titanic Survival Exploration
10. Predicting Boston Housing Prices

Training Models



Training Models

How well is my model doing?

- model good or not
- we are going to learn metrics to tell us if it is good or not

How do we improve the model bases on these metrics?

- we will learn to evaluate and validate and improve

problem

Machine Learning problems

Tools

Regression- Logistic regression, SVM,Naïve….etc.

Measurement Tools

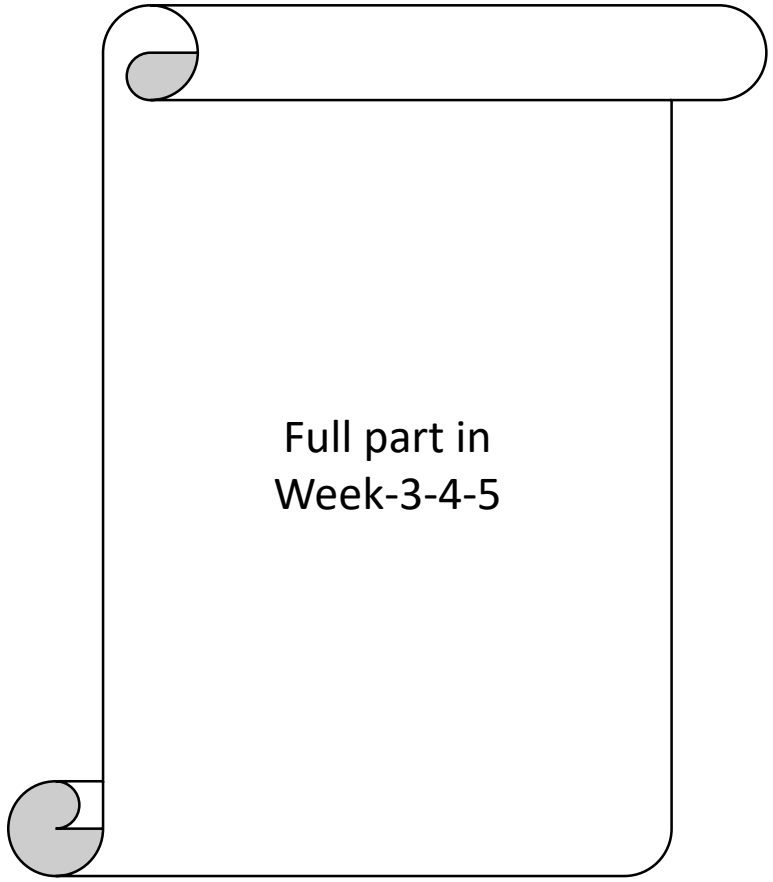
Accuracy, Recall,precesion,F1 score and ….etc.

Statistics Refresher

- we'll assume familiarity with concepts in statistics such as *mean*, *median*, *variance*, etc.

- If it's been a while since you learned these and you feel that you need a refresher, please check our free courses

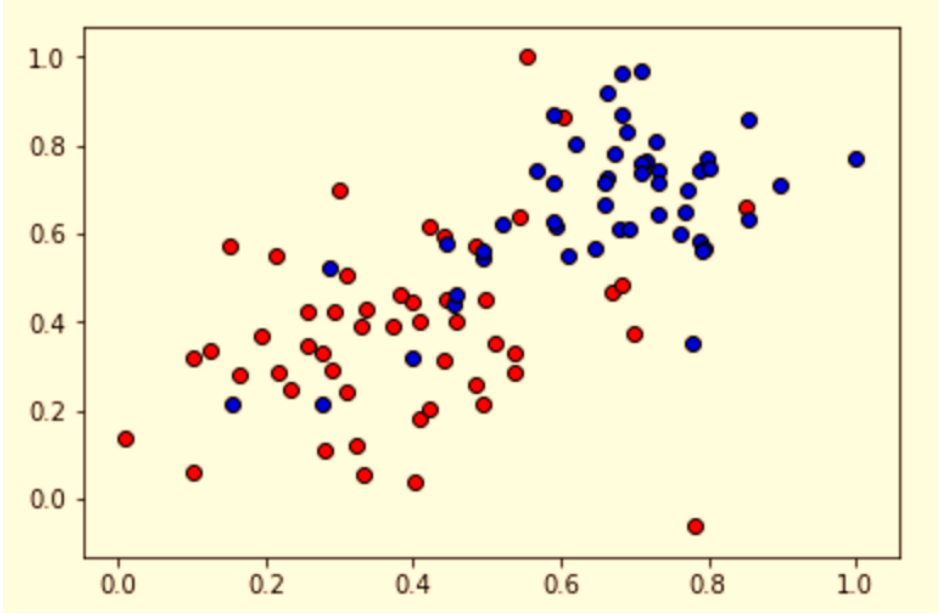
1- Descriptive statistics
2- Inferential statistics.



Pandas

- Loading data into Pandas

- In this section we'll overview the basics on how to open a dataset in Pandas.
- If you want to go more in detail, please check the Pandas Documentation…..we will explain in details later.
- For this, we'll be using the dataset called **2_class_data.csv** which looks as follows:



To open this dataset, which comes as a csv file, we'll use the command read_csv in Pandas:

```
import pandas
data = pandas.read_csv("file_name.csv")
```

Training Models

Numpy

- Now that we've loaded the data in Pandas.
- we need to split the input and output into Numpy arrays, in order to apply the classifiers in scikit learn.
- This is done in the following way: Say we have a pandas data frame called df, like the following, with four columns labeled A, B, C, D:

If we want to extract column A, we do the following:

```
>> df['A']
    0    1
    1    5
    2    9
    Name: A, dtype: int64
```

Now, if we want to extract more columns, we just need to specify them, as follows:

```
>> df[['B', 'D']]
```

And the result is the following Data Frame:

data frame

	A	B	C	D
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

	B	D
0	2	4
1	6	8
2	10	12

And finally, we turn these pandas Data Frames into NumPy arrays. The command for turning a Data Frame df into a NumPy array is very simple:

```
>> numpy.array(df)
```

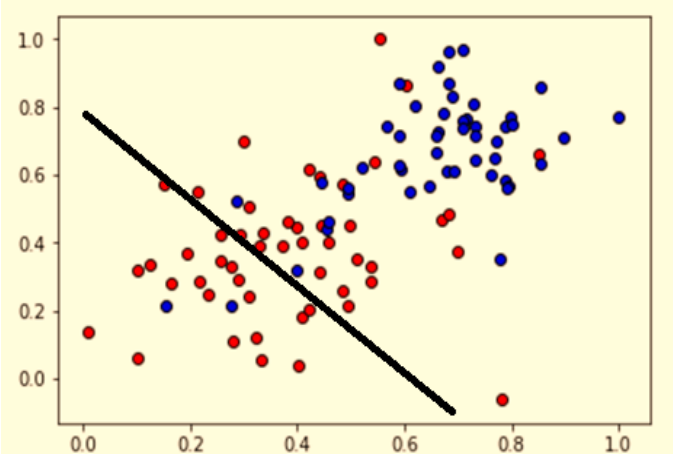
```
import pandas as pd
import numpy as np

data = pd.read_csv("data.csv")

# TODO: Separate the features and the labels into arrays called X and y

X = np.array(data[['x1', 'x2']])
y = np.array(data['y'])
```

scikit learn



most important classification algorithms in Machine Learning, including the following:

- Logistic Regression
- Neural Networks
- Decision Trees
- Support Vector Machines

Now, we'll have the chance to use them in real data! In sklearn, this is very easy, all we do is define our classifier, and then use the following line to fit the classifier to the data (which we call X, y):

Here are the main classifiers we define, together with the package we must import:

```
classifier.fit(X,y)
```

Training Models

Logistic Regression

```
from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression()
```

Neural Networks

```
from sklearn.neural_network import MLPClassifier
classifier = MLPClassifier()
```

Decision Trees

```
from sklearn.tree import DecisionTreeClassifier
classifier = DecisionTreeClassifier()
```

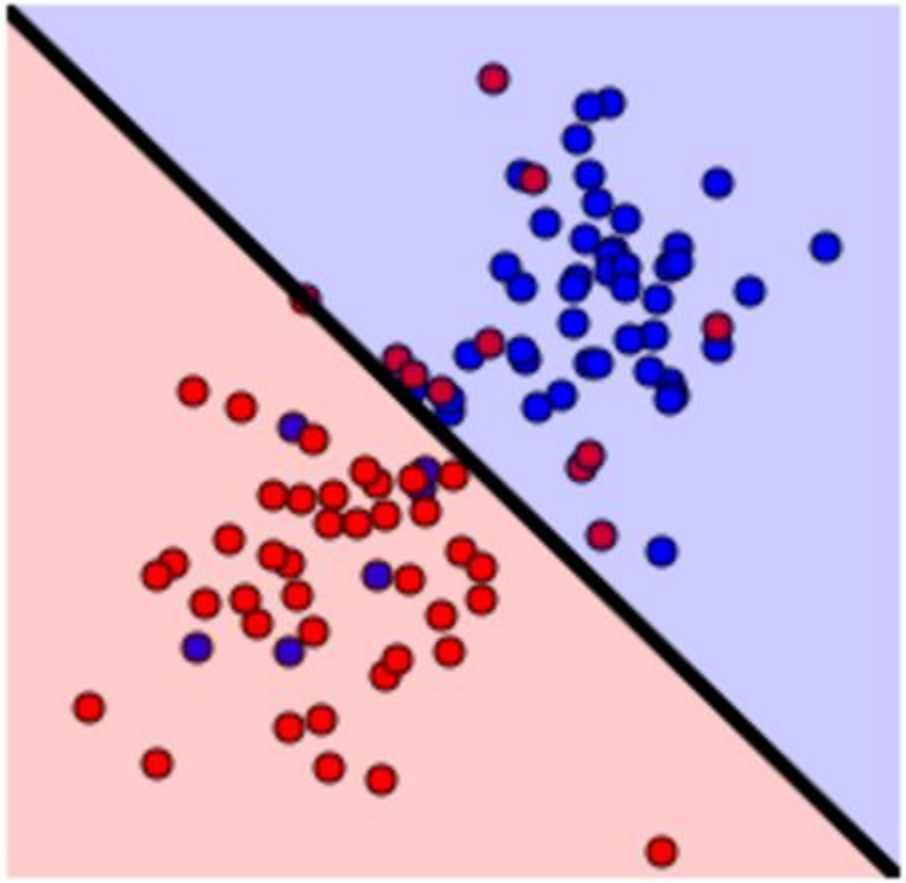
Support Vector Machines

```
from sklearn.svm import SVC
classifier = SVC()
```

Example: Logistic Regression

Let's do an end-to-end example on how to read data and train our classifier. Let's say we carry our X and y from the previous section. Then, the following commands will train the Logistic Regression classifier:

```
from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression()
classifier.fit(X,y)
```



Training Models

Quiz: Train your own model

Your goal is to use one of the classifiers above, between

- ❑ Logistic Regression,
- ❑ Decision Trees,
- ❑ Support Vector Machines
- ❑ Neural Networks are still not available in this version of sklearn, but we will be upgrading soon!), to see which one will fit the data better.

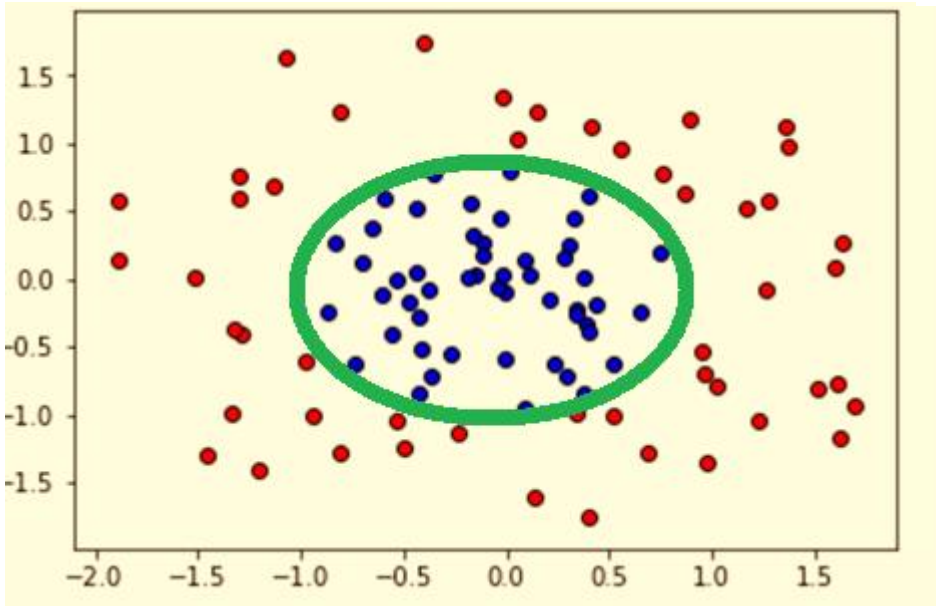
Let's try to fit this data with an SVM Classifier, as follows:

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import GradientBoostingClassifie
from sklearn.svm import SVC

# Logistic Regression Classifier
classifier = LogisticRegression()
classifier.fit(X,y)

# Decision Tree Classifier
classifier = GradientBoostingClassifier()
classifier.fit(X,y)

# Support Vector Machine Classifier
classifier = SVC()
classifier.fit(X,y)
```



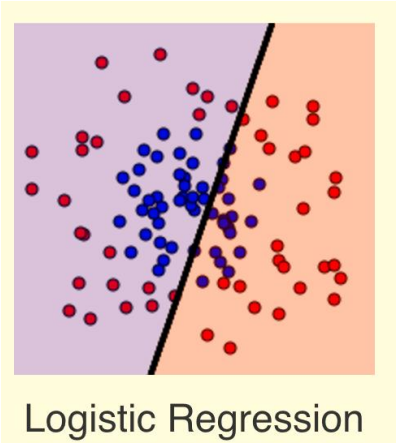
- it seems that maybe we're not exploring all the power of an SVM Classifier. For starters, are we using the right kernel? We can use, for example, a polynomial kernel of degree 2, as follows:

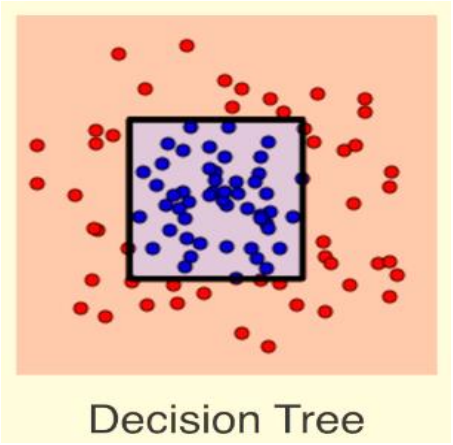
```
>>> classifier = SVC(kernel = 'poly', degree = 2)
```

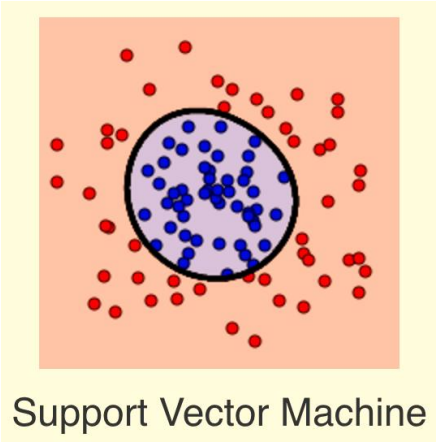
Training Models

Tuning Parameters Manually

So it looks like 2 out of 3 algorithms worked well last time, right? This are the graphs you probably got:



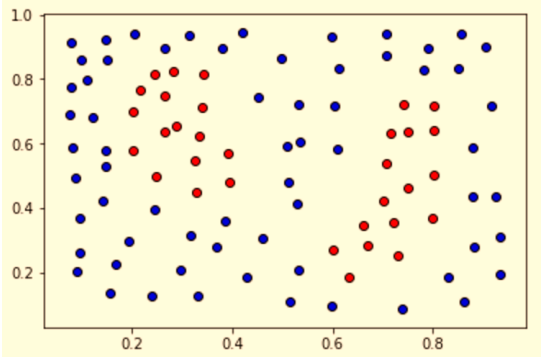




It seems that Logistic Regression didn't do so well, as it's a linear algorithm. Decision Trees managed to bound the data well (question: Why does the area bounded by a decision tree look like that?), and the SVM also did pretty well

Now, let's try a slightly harder dataset, as follows:

```
classifier = SVC(kernel = 'rbf', degree = 2, gamma = 200, C = None)
```



Let's try it ourselves, let's play with some of these parameters. We'll learn more about these later, but here are some values you can play with. (For now, we can use them as a black box, but they'll be discussed in detail during the Supervised Learning Section of this Course.)

- kernel (string): 'linear', 'poly', 'rbf'.
- degree (integer): This is the degree of the polynomial kernel, if that's the kernel you picked (goes with poly kernel).
- gamma (float): The gamma parameter (goes with rbf kernel).
- C (float): The C parameter.

In the quiz below, you can play with these parameters. Try to tune them in such a way that they bound the desired area! In order to see the boundaries that your model created, click on Test Run.

Note: The quiz is not graded. But if you want to see a solution that works, look at the solutions.py tab. The point of this quiz is not to learn about the parameters, but to see that in general, it's not easy to tune them manually. Soon we'll learn some methods to tune them automatically in order to train better models.

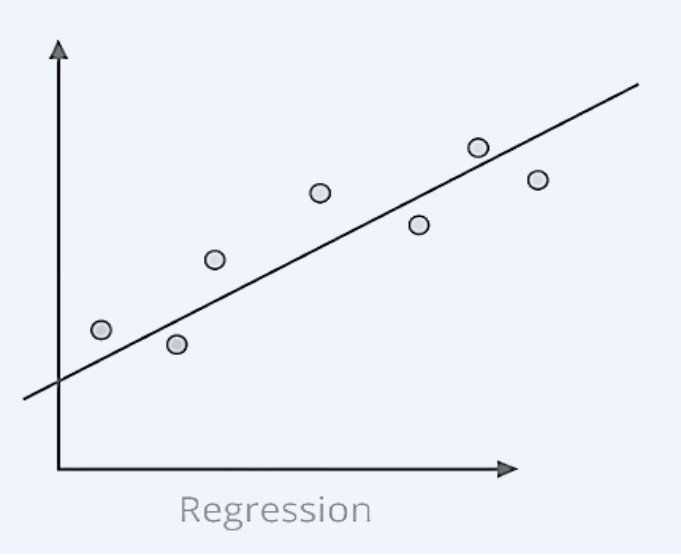
Testing your models

Which Model is better

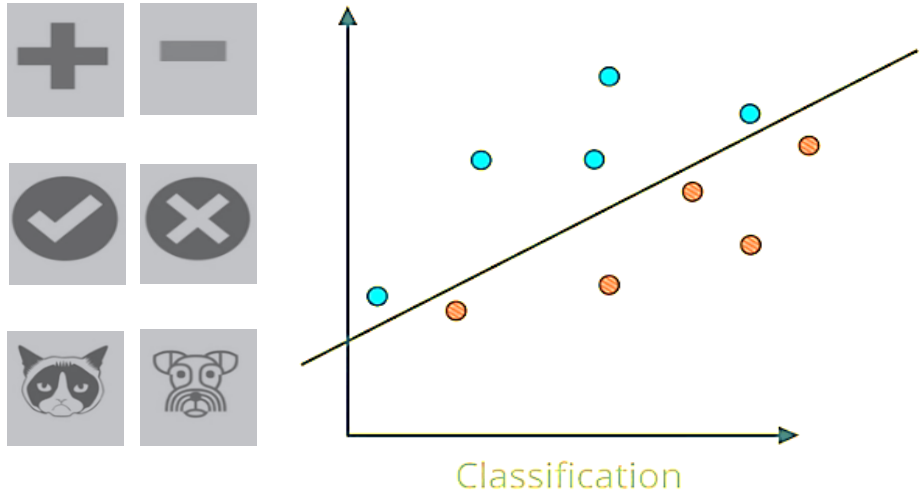
Regression Model : is one that predicts a value like 4 -3 or 6.7 so we need line that fits the data closely if we have anew value on the X axis we approximate by finding the corresponding Y value over the line

Classification Model : when one wants to determine a state such as positive, negative or such as yes, no or cat, dog

Regression return a numeric value



Classification return a state

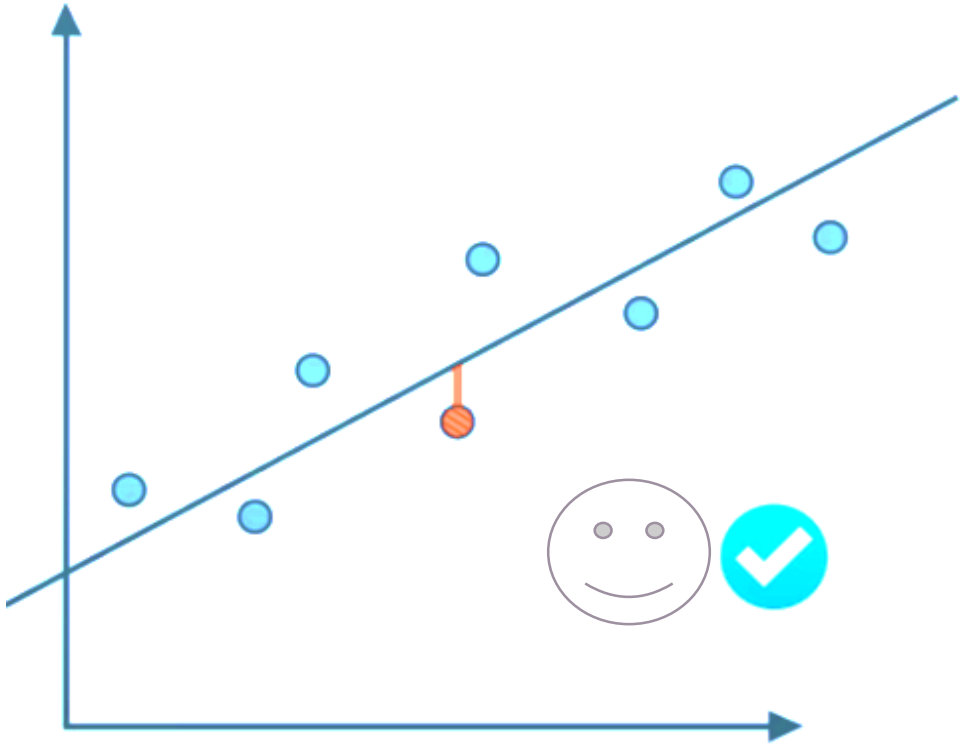


Testing your models

How well is my model doing

Which Model is better

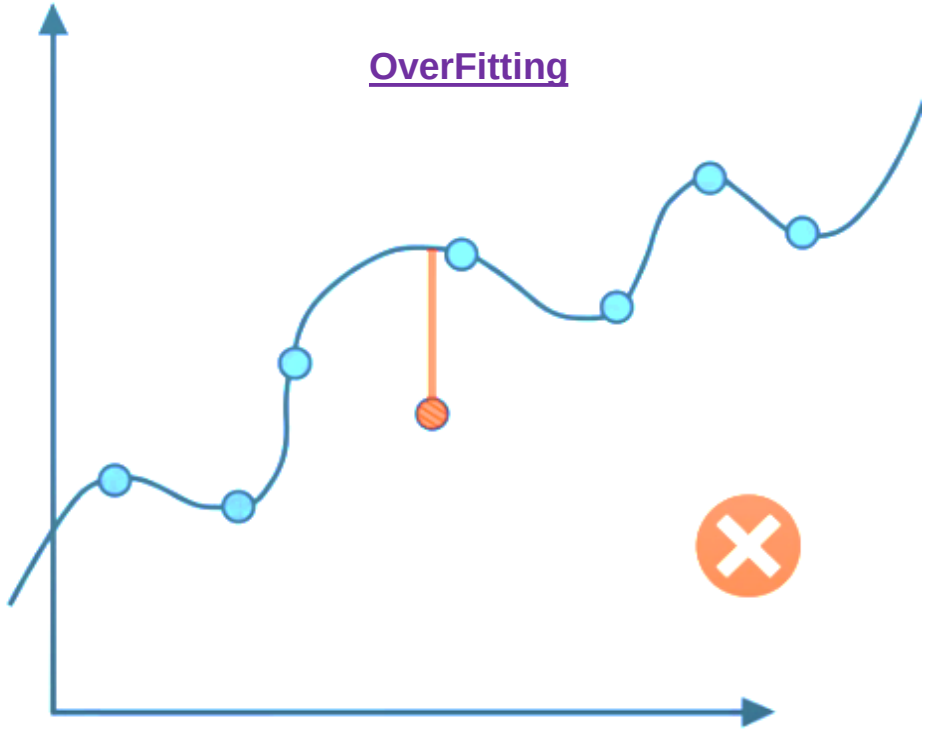
Des not fit the data



How do we find a model that generalizes well?

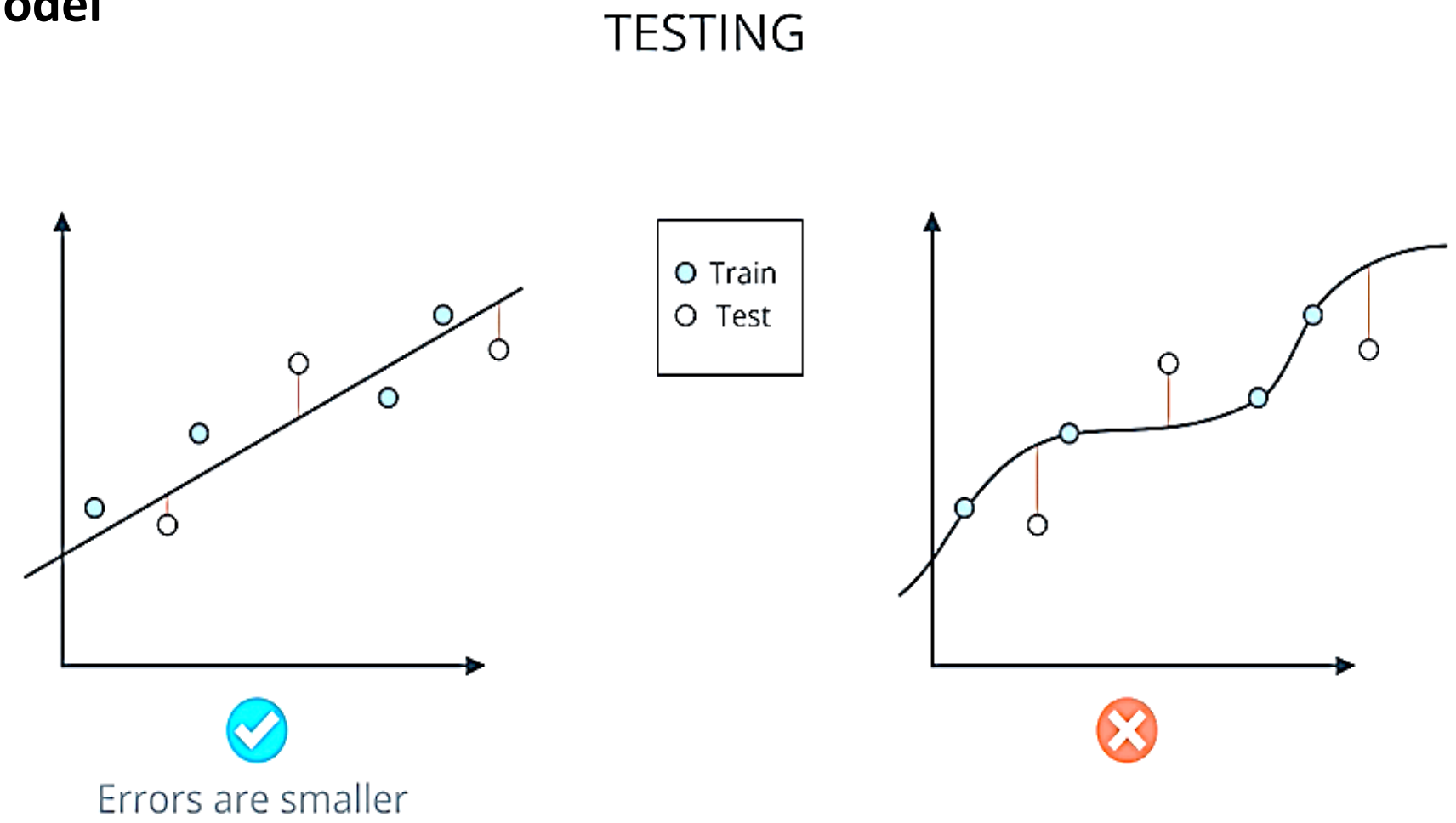
Fit the data

OverFitting

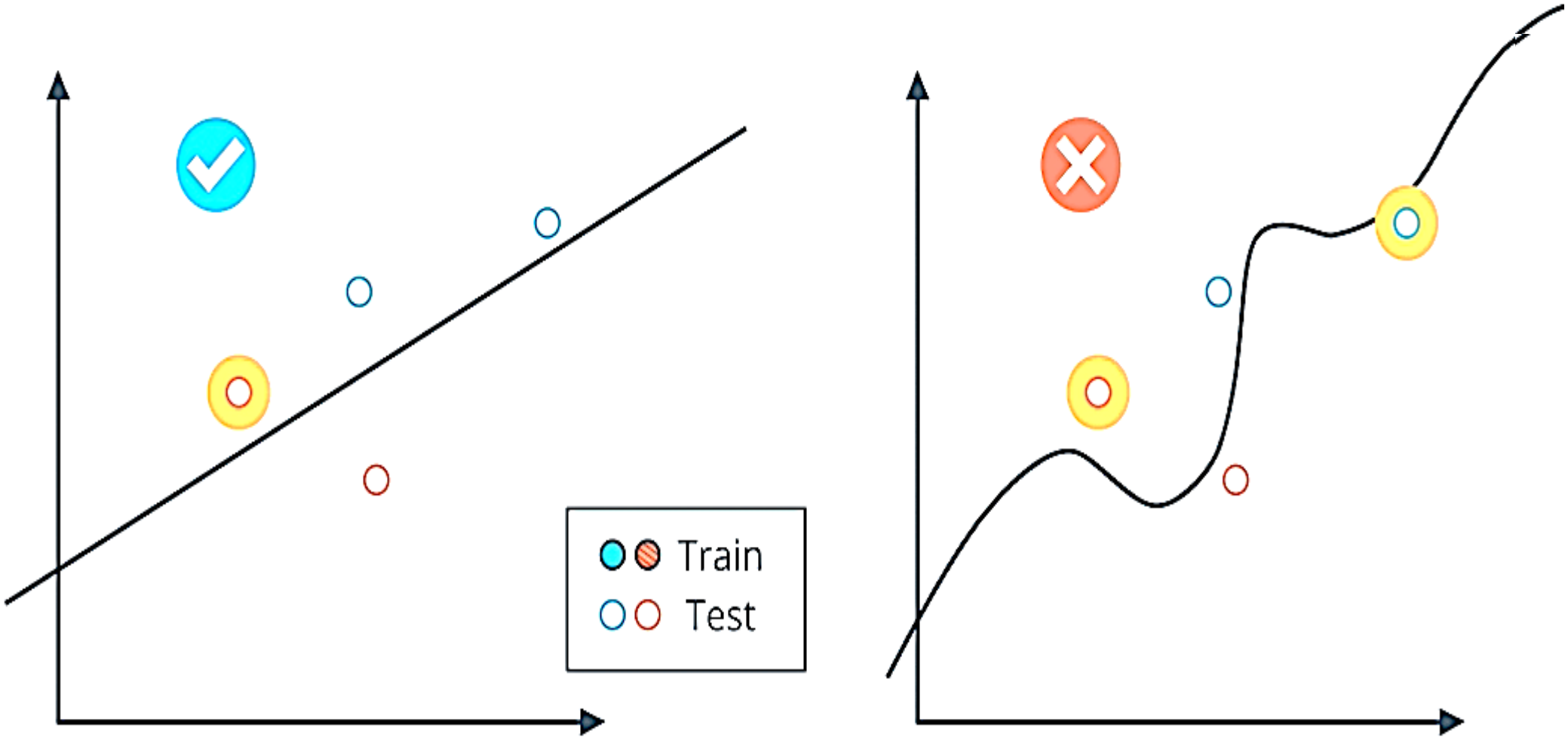


The answer of this question …this is where we introduce the concept of testing.

Regression Model



Classification Model

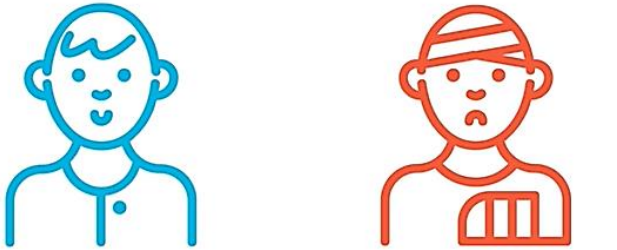


Testing in SKlearn

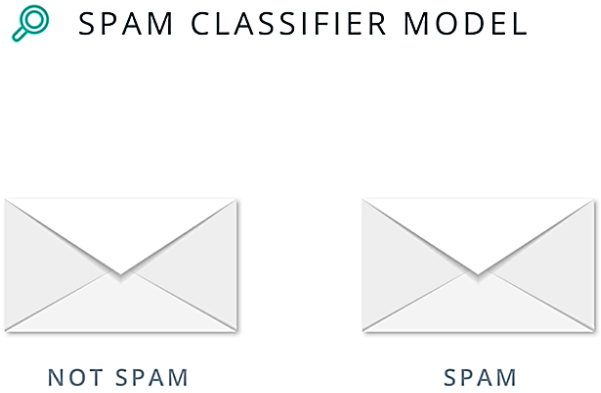
Evaluation Matrix

How well is my model doing?

Example-1
Medical Model



Example-2



Evaluation Metrics

MEDICAL MODEL

When a patient is healthy and the model incorrectly diagnoses them as sick, this is also a mistake and it means we will be sending a healthy person for further examination or treatment

	Diagnosed Sick	Diagnosed Healthy
Sick	 True Positive	 False Negative
Healthy	 False Positive	 True Negative

Evaluation Metrics

- Model that will help us detect a particular illness and tell if a patient is healthy or sick

10, 000 PATIENTS

CONFUSION MATRIX

DIAGNOSIS

PATIENTS	Diagnosed Sick	Diagnosed Healthy
Sick	1000 True Positives	200 False Negatives
Healthy	800 False Positives	8000 True Negatives

Evaluation Metrics

- Model that will help us detect a spam detector which will help us determine if an email is spam or not

SPAM CLASSIFIER MODEL

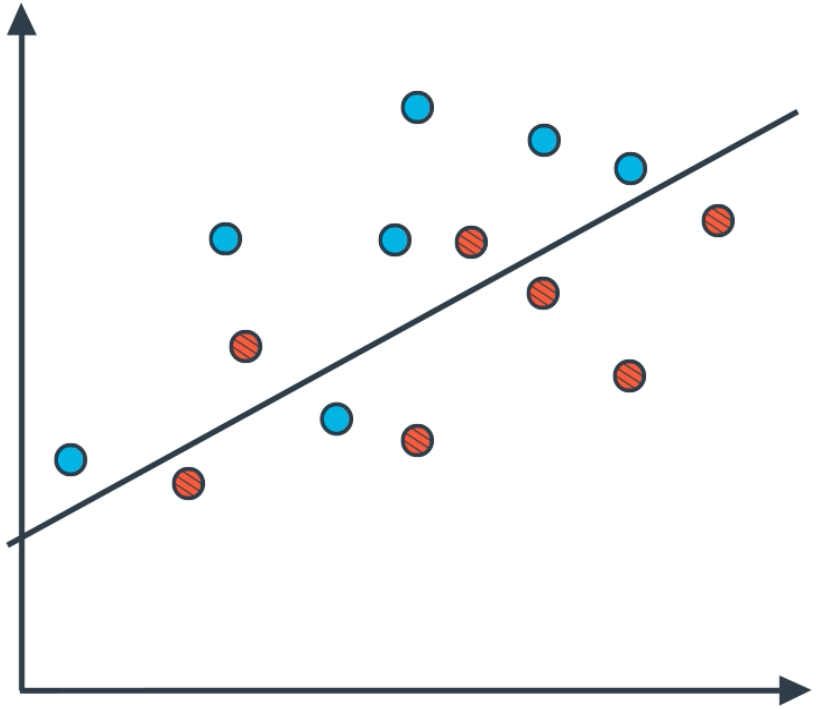
CONFUSION MATRIX

1000 EMAILS

EMAIL	SPAM: Spam Folder	SPAM: Inbox
Spam	100 True Positives	170 False Negatives
Not Spam	30 False Positives	700 True Negatives

Confusion Matrix Quiz

CONFUSION MATRIX



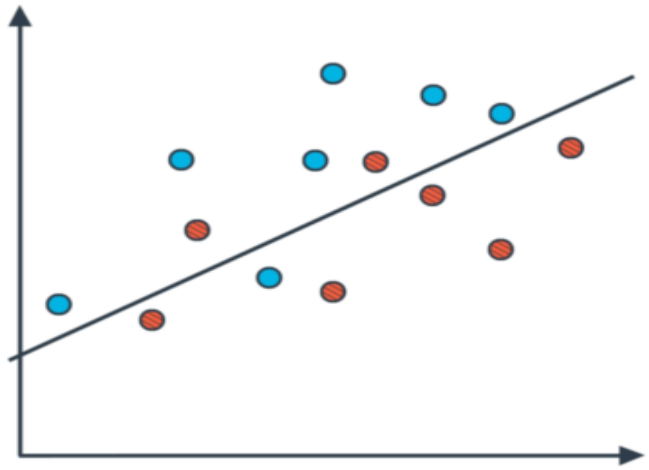
	Guessed Positive	Guessed Negative
Positive	6 True Positives	1 False Negatives
Negative	2 False Positives	5 True Negatives

How many True Positives, True Negatives, False Positives, and False Negatives, are in the model above?

Evaluation Metrics

Type 1 and Type 2 Errors
Sometimes in the literature, you'll see False Positives and False Negatives as Type 1 and Type 2 errors. Here is the correspondence:

Type 1 Error (Error of the first kind, or False Positive): In the medical example, this is when we misdiagnose a healthy patient as sick.
Type 2 Error (Error of the second kind, or False Negative): In the medical example, this is when we misdiagnose a sick patient as healthy.



	Guessed Positive	Guessed Negative
Positive	6 True Positives	1 False Negatives
Negative	2 False Positives	5 True Negatives

Accuracy

How many did we classify correctly?

One of the ways to measure how good a model is .

The answer is, the ratio between the number of correctly classified points and the number of total points.

	Spam folder	Inbox
Spam	100	170
Not spam	30	700

Medical model

$$\text{Accuracy} = \frac{1000 + 8000}{10000} = 90\%$$

Spam detector model

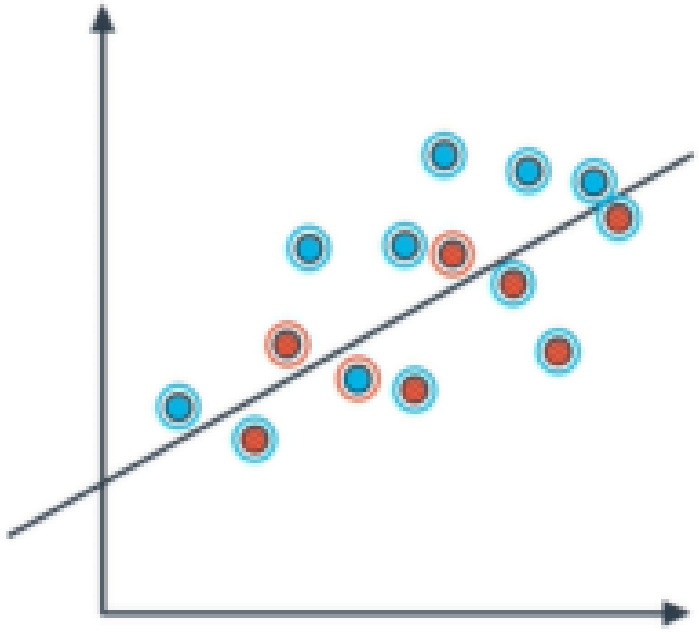
$$\text{Accuracy} = \frac{100 + 700}{1000} = 80\%$$

```
From sklearn.metrics import accuracy_score
```

Quiz

- ACCURACY

Out of all the data, how many points did we classify correctly?



$$\text{Accuracy} = \frac{\text{Correctly classified points}}{\text{All points}}$$

$$= \frac{11}{11+3} = \frac{11}{14} = \mathbf{78.57\%}$$

It's pretty tricky to just look at accuracy and use that to evaluate our model because it may completely miss the point when the data is skewed like this one.

When accuracy won't work?

Credit card fraud

Good

284335 Good transactions

Fraudulent

472 Fraudulent transactions

Accuracy ???

$$\text{Accuracy} = \frac{284335}{284887} = 99.83\%$$

This model must be pretty good if it's accurate is that high right?????

Try another model

Not really

This model is not catching any of the bad ones and the point of the model Is to catch the fraudulent transactions

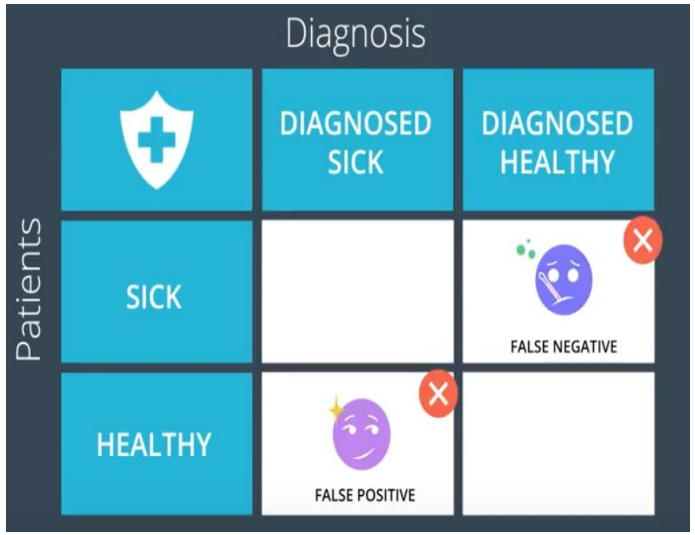
It's accidentally catching all the good ones.

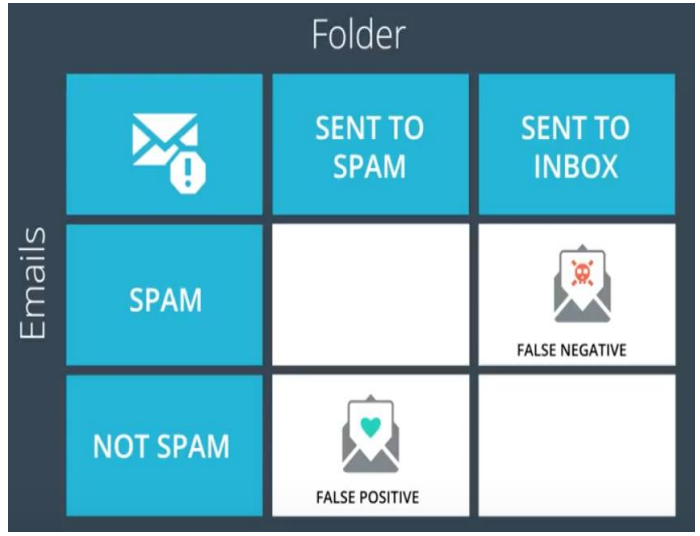
Terrible model

Can we get a model that catches all the bad transaction?

I'm going to label all transactions fraudulent. So, that's great….right?

What be worst? 🤔 **False Negatives and Positives**





Quiz 1: The Medical Model

Question

In the medical example, what is worse, a False Positive, or a False Negative?

False Positives

✓ **False Negatives**

Quiz 2: The Spam Model

Question

In the spam detector example, what is worse, a False Positive, or a False Negative?

✓ **False Positives**

False Negatives

Precision and Recall

Solution: False Positive and Negative

So this model basically says "i don't really care if i find all the spam emails but one thing is for sure
if i say that an email is spam it better be a spam"

- A false negative means a spam email made its way into your inbox and all you had to do is delete it.

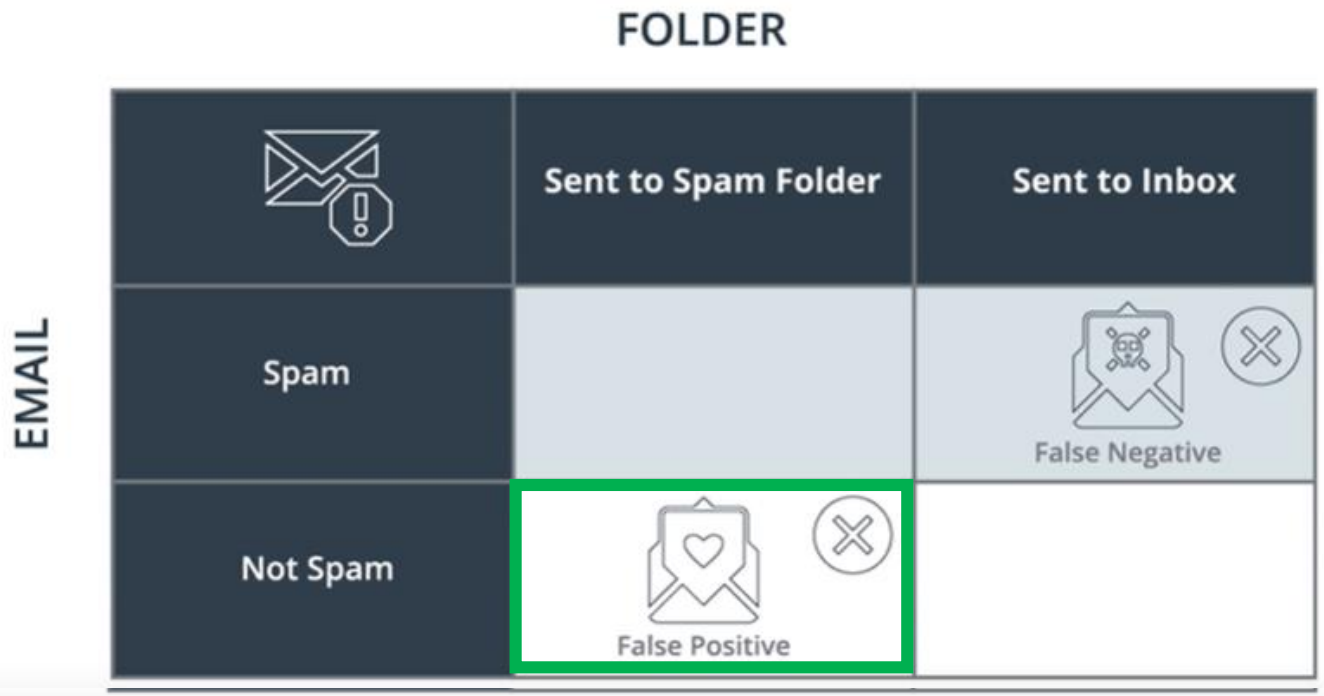
So it's a bit of an inconvenience

- On the other hand what's a false positive?

Your grandma the poor lady learn to type an email only to tell you she bake cookies and you delete it .

that's terrible

So here we want to punish false positives more than false negatives



False negative is much worse than a false positive, since predicting that a sick person is healthy is much more dangerous than predicting that a healthy person is sick.

Medical Model

FALSE POSITIVES OK

FALSE NEGATIVES NOT OK

OK IF NOT ALL ARE SICK
FIND ALL THE SICK PEOPLE

For this we introduce two metrics,
<u>precision and recall</u>

they will measure the exact things that we want ,

- **the medical model need to be a high recall model**
- **the email model needs to be high precision model**

Spam Detector

FALSE POSITIVES NOT OK

FALSE NEGATIVES OK

DON'T NECESSARILY NEED
TO FIND ALL THE SPAM
BETTER BE SPAM

we can see the medical model and the email model are fundamentally different
the medical model as we saw is okay with false positive but not with false negative
since we're okay misdiagnosing some patient as sick as long as we find all the sick people

the email model is okay with false negative but not with false positives since we don't necessarily need to find all the spam emails but one thing is for sure that if we do label something as spam it better be spam.

Precision

so precision is this column because this column are the sick patient that we diagnose as sick

$$precision = \frac{1000}{1000+800}$$

$$=55.6\%$$

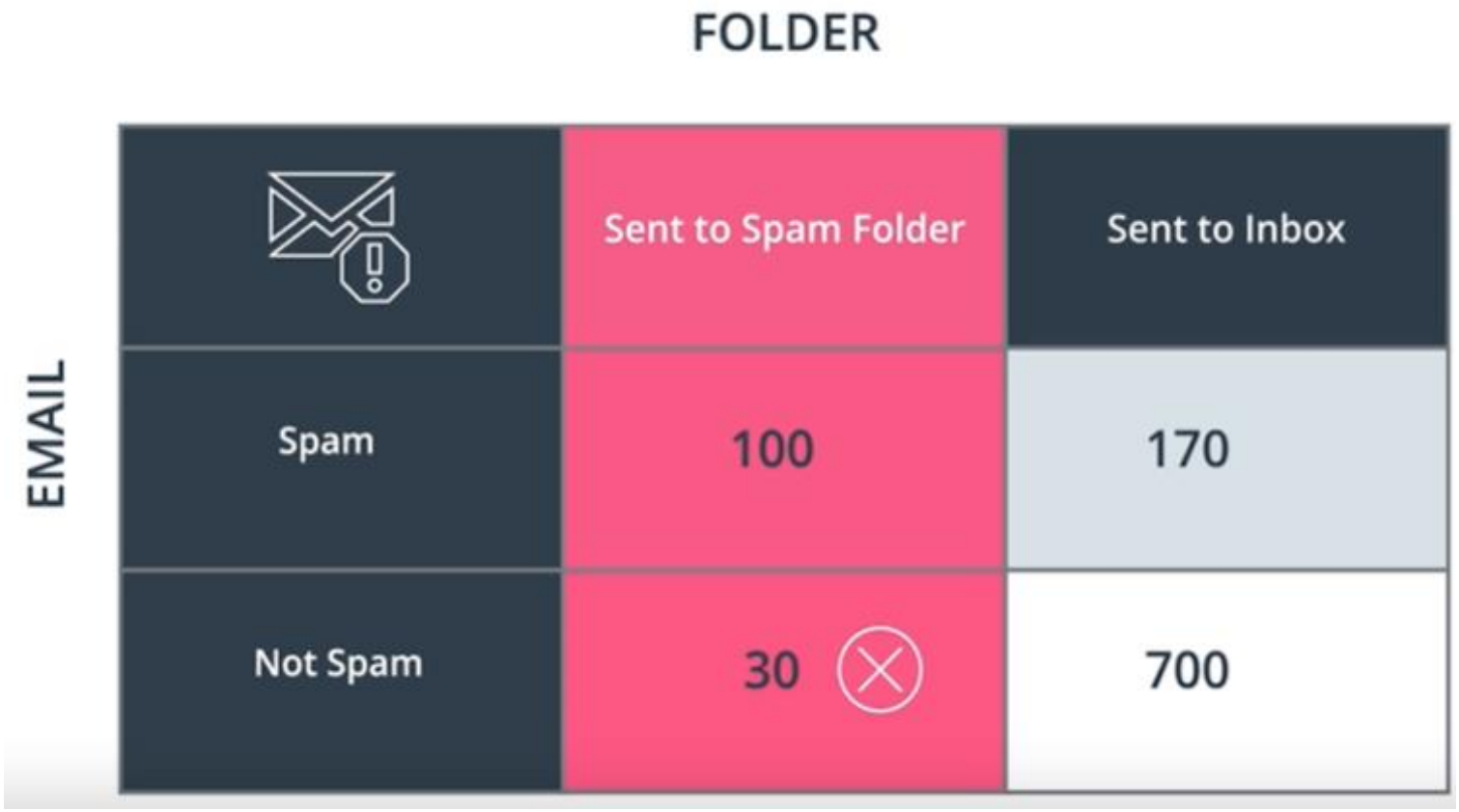
$$precision = \frac{TP}{TP+FP}$$

Out of all the points predicted to be positive How many of them were actually positive?

Out of all the patients that we diagnosed as sick. How many were actually sick?

Precision

Spam email model

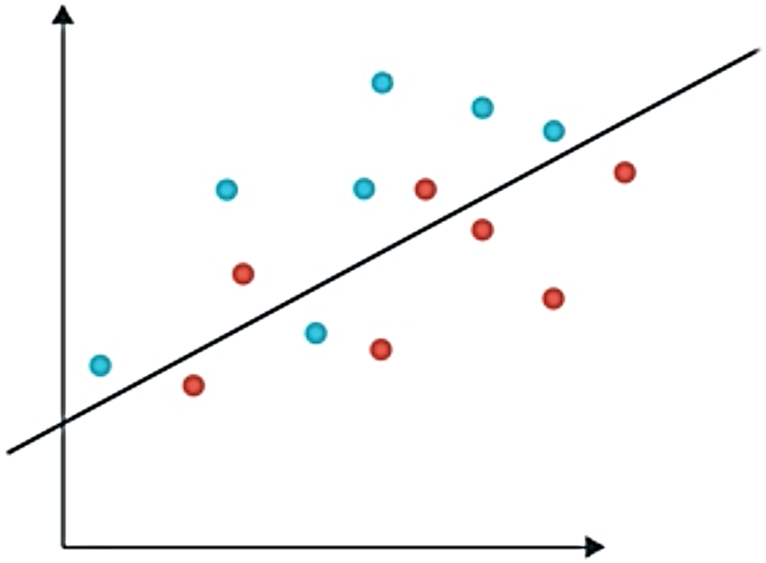


$$\text{precision} = \frac{100}{100+30} = 76.9\%$$

Out od all the Emails….sent to the spam folder ----------how many were actually spam?

Precision Quiz

- PRECISION



OUT OF THE POINTS WE HAVE PREDICTED TO BE POSITIVE, HOW MANY ARE CORRECT?

FOLDER

EMAIL	Sent to Spam Folder	Sent to Inbox
Spam	True Positive	False Negative
Not Spam	False Positive	True Negative

$$\text{precision} = \frac{6}{6+2} = 0.75$$

Recall

Out of the points that are labeled positive, How many of them were correctly predicted as positive?

For medical model

How many did we correctly diagnose as sick?

Very important measure on the medical field and they called as sensitivity.

DIAGNOSIS

PATIENTS	Diagnosed Sick	Diagnosed Healthy
Sick	1000	200
Healthy	800	8000

How many of the positive points did I manage to catch?

Recall
Is opposite
Out of the patients that are sick, How many did we correctly diagnose as sick?

$$\text{Recall} = \frac{1000}{1000+200} = 83.3\%$$

Recall

For Email model

How many of them do we correctly send to spam folder?

Remember

We are worried about avoiding this X over here since we don't mind if we don't catch all the spam emails as long as the ones we caught are spam

FOLDER

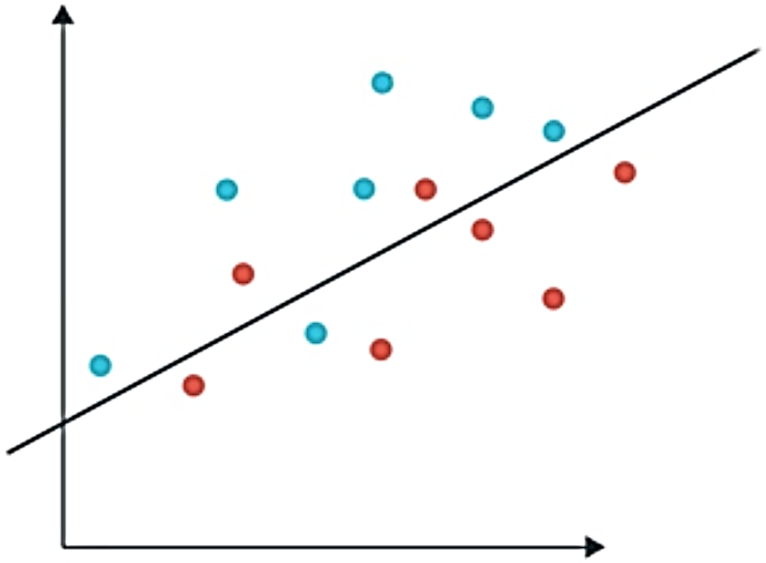
EMAIL	Sent to Spam Folder	Sent to Inbox
Spam	True Positive	False Negative
Not Spam	False Positive	True Negative

$$\text{Recall} = \frac{100}{100+170} = 37\%$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall Quiz

- PRECISION



OUT OF THE POINTS WE HAVE PREDICTED TO BE POSITIVE, HOW MANY ARE CORRECT?

EMAIL	FOLDER: Sent to Spam Folder	FOLDER: Sent to Inbox
Spam	6	1
Not Spam	2	5

$$\text{Recall} = \frac{6}{6+1} = 0.857$$

F1 Score

- F_1 SCORE

ONE SCORE?

MEDICAL MODEL

PRECISION: 55.7%

RECALL: 83.3%

AVERAGE=69.5%

SPAM DETECTOR

PRECISION: 76.9%

RECALL: 37%

AVERAGE=56.95%

Good Metric but I'm sure you're probable thinking in your head!!!!!!!!!!

that's not very different than accuracy that's not saying too much.

And the way to see how this average is not the best idea is to try it in the extreme example

The question is

Do we want to be carrying two numbers around all the time?

Do we want to be carry precision in one pocket and recall in the other one?

We kind of want to only have one score.

And I will be looking at the both any time we make a decision.

Again the question is ……How do we combine these two scores into one?

Can we think of a way?
Well, a pretty simple ways taking the average of precision and recall

F1 Score

The question is
What is the precision and Recall of this model?

Precision=100%

Average=50%

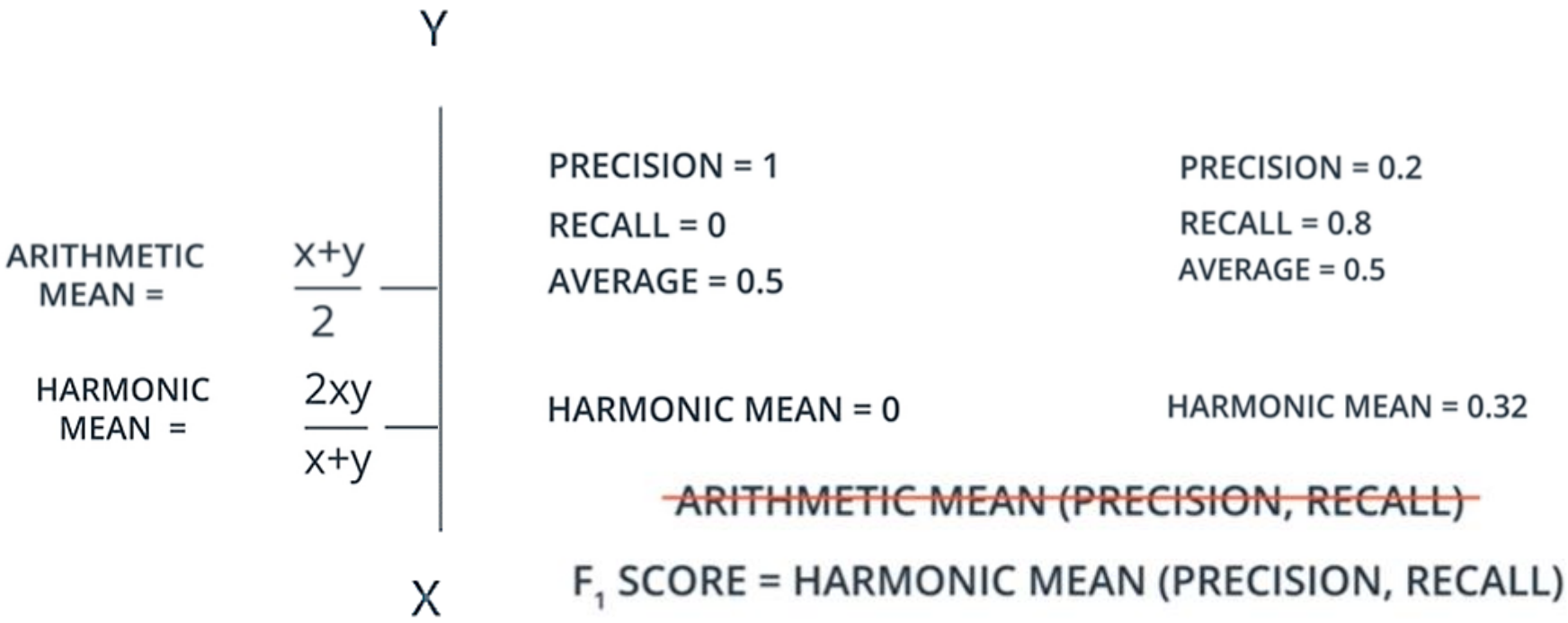
Recall=0%

AVERAGE = 50.08%

PRECISION = 472/284,807 = 0.16%

F1 Score

- HARMONIC MEAN



Precision Quiz

If the Precision of the medical model is **55.6%**, and the Recall is **83.3%**, what is the F1 Score?